

Simulation Study on Variable Screening Methods in High-Dimensional Logistic Regression

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April 18, 2025

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1 Introduction

In modern data science applications, it is easy to see datasets often contain an enormous number of predictors relative to a considerably smaller set of observations. Presently in areas such as genomics, image processing, finance, and natural language processing, data scientists have to constantly deal with datasets in ultrahigh dimensionality. This has led to the emergence of NP-dimensionality statistical problems, where $p \gg n$. Being able to handle NP-dimensional problem yields significant real-world benefits: identifying disease markers in gene profiles, screening vast libraries of drug compounds, constructing optimal investment portfolios, etc.

The traditional techniques such as OLS break down in NP-dimensional setting due to overfitting, multicollinearity, and computation instability. To learn more of the dataset and their responses, the challenge is not only fitting the model, but it is also to select the subset of variables that is truly relevant in the model among countless irrelevant ones. Generally, selecting only relevant predictors is hyper-crucial for improving model interpretability, reducing overfitting, and making it less computationally expensive. This sparks the need for robust variable selection approaches, as even established penalized methods like LASSO and SCAD can face challenges under certain high-dimensional conditions.

Thus, it led to the first introduction of Sure Independence Screening (SIS) by Fan & Lv (2008) for linear models. The new screening method is simple, easy-to-implement, and fast, which drastically reduce dimensionality by ranking features based on the correlation. This helps retain the variables with high probability, hence sure screening property. However, the original SIS witnessed several limitations: it is heavily predicated on normality assumption and it is not easily extensible even in linear regression settings. The aforementioned points culminated to the introduction of SIS for Generalized Linear Models by Fan & Song in 2010, which aims to improve the vanilla SIS framework by introducing variable screening by ranking Maximum Marginal Likelihood Estimator magnitude (SIS-MMLE) and by Marginal Likelihood Ratio (SIS-MLR). While iterative extensions like ISIS exist to boost performance, this report solely focuses on evaluating the properties of the basic SIS procedures as presented by Fan & Song (2010).

This report aims to replicate and evaluate the performance of these SIS methods under logistic regression in setting S3 from Fan & Song (2010). Under the same conditions, the extended SIS is then compared against LASSO and SCAD using two metrics Minimum Model Size (MMS) and Robust Standard Deviation (RSD) as similarly presented in the paper. This setting was chosen specifically for its complex correlation structure, known to be challenging for methods like LASSO and SCAD.

This report proceeds by first reviewing background concepts, detailing further the concepts of GLMS, penalized selection methods, and the SIS framework (Section 2). In section 2, I will discuss more about the metrics used to evaluate selection methods' performances. Section 3 details the simulation setup used to replicate Setting S3 and specifically Table 5 from Fan & Song (2010). The performance of different screening and selection methods is presented and evaluated based on the recorded simulation results in Sections. Finally, Section 5 concludes the report by providing summary points, and their implications.

2 Background

Generalized Linear Models (GLMs) is an extension of linear regression for response variables with different distributions like binomial or Poisson which is more flexible than the ordinary least squares regression presented in Fan & Lv (2008). In this report, I am only focusing on logistic regression model to predict if the response variable $\mathbf{Y}$ will fall into one of its two possible categories based on predictor variables $\mathbf{X}$. This can be achieved using `logLik()` function for binomial error distribution and `glm()` with `family=binomial` for the logistic regression model itself.

In Fan & Song (2010), NP-dimensionality problem is represented as $\log(p) = O(n^a)$, where p is the number of predictors, n is the number of observations with a dimensions. This is also how the dimensionality problem is presented within the report. Moreover, the key assumption in this report and in the paper is the sparsity in the feature space, where only a small subset of p predictors truly influences the response variable. Model $\mathcal{M}_*$ is set up in a way that if β^* is the true vector of regression coefficients, the set of important predictors becomes $\mathcal{M}_* = \{1 \leq j \leq p : \beta_j^* \neq 0\}$. This indicates that the number of truly important variables is drastically smaller than the total number of variables, where $s \ll p$ and $s = |\mathcal{M}_*|$.

To highlight the performance of SIS in this report, it is important to take in account two traditional penalized variable selection methods which were proven to be highly effective in similar model configuration. LASSO or Least Absolute Shrinkage and Selection Operator was introduced by Tibshirani (1996), which uses $\mathcal{L}_1$ ($\lambda \sum |\beta_j|$) to perform continuous coefficients shrinkage for variable selection. On the other hand, SCAD or Smoothly Clipped Absolute Deviation, proposed by Fan & Li (2001), uses non-convex penalty to reduce estimation bias for $\mathcal{L}_1$ penalty for large coefficients. These methods are accessible via `glmnet` and `ncvreg` packages in R, which are both utilized in this report. Albeit these methods are powerful under certain conditions, they can fail when the predictors are highly correlated or when specific theoretical requirements are violated, especially for LASSO.

This report main goal is to highlight the performance of Sure Independence Screening (SIS) in variable selection, which is the first step to solve ultrahigh-dimensional data problem. SIS operates as a fast system that predicates on marginal screening. It ranks all p variables based on their individual link with the response. This effectively filters out the variables with weak marginal utility. The goal of SIS is to reduce the dimensionality from p to a more manageable size d , where it is often that $d < n$. The theoretical property makes SIS effective and correct is the sure screening property. This means that under certain condition, SIS guarantees to capture a set of variables that contain the true set of active predictors $\mathcal{M}_*$, and the probability of this property increases as the sample size increases. In this report, MMLE and MLR are used to enhance the performance of SIS. MMLE or Maximum Marginal Likelihood Estimate fits p separate simple GLMs, which involves only the intercept and one predictor. It then ranks the predictors based on the absolute value of the estimated marginal coefficients. The variables with larger coefficients are considered more important. This method is implemented inside the function `evaluate_sis_mmle()` using R SIS package. Another method used to support SIS variable selection is MLR or Marginal Likelihood Ratio. Using simple marginal GLMs fitting scheme as MMLE, the predictors are now ranked based on the improvement in likelihood over the null model. This is manually implemented in the `evaluate_sis_mlr()` function.

The evaluation processing this report is based on two metrics: Minimum Model Size (MMS). MMS is defined as the smallest integer k such that the set of top k ranked predictors includes all s truly active predictors $\mathcal{M}_*$. A smaller MMS indicates better performance since it requires less work to include all important variables. Also, smaller model increases stability and interpretability. When the method fails to include all true predictors even after selecting a large number of variables, its MMS is recorded as the total number of predictors p . While SIS limit its variable selection up to $d = nsis$ steps, LASSO and SCAD requires full-path run to determine. Robust Standard Deviation (RSD) is also used to examine data variance, calculated as IQR divided by 1.349.

3 Simulation Design

This project aims to repeat one set of the simulations of Fan & Song (2010), which evaluated the performance of the SIS framework using the Marginal Maximum Likelihood Estimator and Marginal Likelihood Ratio, LASSO, and SCAD under high-dimensional logistic regression. In the section below, I will target setting S3 which was reported in table 5 of the paper. The simulation conducted in section below aims to replicate the findings in the original paper, while introducing a few tweaks.

The specifications of the simulation setup are as follows:

- Sample size: $n = 600$
- Number of predictors: $p = 2000$
- Number of true predictors: $s \in \{3, 6, 12, 24\}$
- Number of simulations: 200

To ensure the similar simulation setting, I replicated data generation through the specifications given in the paper. First, I generated a data matrix $\mathbf{X}$ with $p-50$ predictors which are identically and independently distributed by $\mathcal{N}(0, 1)$. Explicitly, the last 50 predictors of $\mathbf{X}$ are generated as $X_k = \sum_{j=1}^s X_j(-1)^{j+1}/5 + \sqrt{25-s}/5\epsilon_k$, where X_j refers to the first s predictors and ϵ_k follows $\mathcal{N}(0, 1)$. The goal of parsing the covariates in the first s predictors and also the last 50 is to introduce a more complex variable selection task. Then, all predictors in $\mathbf{X}$ are standardized to have a mean of 0 and variance 1 using `scale()`. I used a special metric called True Model (β^*), which defines the true coefficient vector β^* representing the true predictors located among the last 50 variables in $\mathbf{X}$. Finally, I produced response $\mathbf{Y}$ from $\mathbf{X}$ from a Bernoulli distribution with probability function $P(Y = 1|X) = \frac{\exp(X\beta^*)}{1 + \exp(X\beta^*)}$, using the `rbinom()`. Furthermore, it is preferred in the paper that one chooses $|\mathcal{M}_{\lambda_n}|$ or $n/\log(n)$ as their sufficient number of variables for SIS variable selection. Hence, in this report, SIS-MMLE and SIS-MLR were enforced with this threshold, while MMLE-Rank was designed to forgo this constraint to test whether SIS better performed in the specified setting without such limit.

Regarding the choice of true coefficient vector β^* , they were constructed according to the sparsity statistic s . Because the β coefficients were not specified for table 5 in Fan & Song (2010), I decided to design this vector based on the ones from table 2 to be more aligned with the original paper. The coefficients were set to zero except for indices $j = p - 50 + 1, \dots, p - 50 + s$, which are the first s predictors of the last 50 variables. The magnitudes of the important variables were as follows:

- For $s = 3$: $\beta_{active} = (3, 4, 3)$
- For $s = 6$: $\beta_{active} = (3, -3, 3, -3, 3, -3)$
- For $s = 12$: $\beta_{active} = (3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4)$
- For $s = 24$: $\beta_{active} = (3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4, 3, 4)$

For each simulated data set, the following methods were applied and their Minimum Model Size (MMS) were recorded. A result of $p = 2000$ indicates failure to recover important information within the model. Plus, the maximum eigenvalue $\mathcal{M} \cdot \lambda_{max}$ of the sample covariance matrix was calculated to examine further the model stability.

- $\mathcal{M} \cdot \lambda_{max}$: For each generated $\mathbf{X}$ matrix, sample covariance matrix was computed using `cov(X)`. Then, its eigenvalues were retrieved via `eigen()` and the maximum eigenvalue was recorded. The median and RSD is also recorded, matching closely to the definition from Fan & Song (2010).

- **MMLE-Rank**(`evaluate_mmle_rank`): This method is manually implemented as a variation of SIS-MMLE. All p predictors were ranked based on the absolute magnitude of their marginal logistic regression coefficients. In specifics, the coefficient is estimated by fitting $Y \sim X$ using `glm(family=binomial)`. MMS was recorded as the smallest number k where $k \in 1, \dots, p$ such that all true variables were included.
- **SIS-MLR**(`evaluate_sis_mlr`): This method is manually implemented because **SIS** package does not include an option for MLR as presented in the paper. The marginal likelihood ratio statistic was calculated for each predictor by comparing the likelihood marginal model ($Y = X_j$) to the null model ($Y = 1$) using `glm` and `logLik` functions. The top $nsis = \frac{n}{\log n} = 93$ predictors were selected. MMS was calculated as the smallest k within the set required to include all relevant variables.
- **SIS-MMLE**(`evaluate_sis_mmle`): This implementation utilized the **SIS** function from the same name package. The function performs an initial screening and returns indices of the top $nsis$ variables. MMS was calculated as the smallest k within the set required to include all relevant variables.
- **LASSO** (`evaluate_glmnet`): LASSO was implemented using `cv.glmnet` from the **glmnet** package with `alpha=1` and 10-fold cross-validation. The regularization parameter λ was chosen using "one standard error" rule i.e `s = "lambda.1se"`. MMS was reported as the number of non-zero coefficients if all true variables were selected, otherwise p .
- **SCAD** (`evaluate_ncvreg`): SCAD was implemented using `cv.ncvreg` from the **ncvreg** package with `penalty="SCAD"`, $\gamma = 3.7$, and 10-fold cross validation. The regularization parameter λ was chosen to minimize cross-validation error i.e `lambda = cv_fit$lambda.min`. MMS was reported as the number of non-zero coefficients if all true variables were selected, otherwise p .

Nonetheless, `set.seed(123456789)` was utilized for results reproducibility. The simulation number was set to 200, which was specified in the paper. Using `for-loop`, the final results were appended after each iteration by calculating the MMS and RSD across 200 simulations for each value of s . The format of the output was targeted to resemble the table 5 in Fan & Song (2010).

```
cat("Simulations complete.\n\n")

# print descriptive headers for the output table
cat("TABLE 5 (Reproduced - Setting S3)\n")
cat("The Median MMS / M-lambda_max and the associated RSD (in parentheses)\n")
cat("Setting 3 (S3), p =", p, ", n =", n, ", num_simulations =", num_simulations, "\n\n")

# print headers
header <- sprintf("%-3s | %-16s | %-17s | %-15s | %-15s | %-15s | %-15s\n",
  "s", "M-lambda_max(RSD)", "MMLE-Rank(RSD)", "SIS-MLR(RSD)", "SIS-MMLE(RSD)", "LASSO(RSD)", "SCAD(RSD)")
cat(header)
cat(rep("-", nchar(header)), "\n", sep = "")

# print results
for (i in 1:nrow(results)) {
  row <- results[i, ]
  cat(sprintf("%-3d | %-16s | %-17s | %-15s | %-15s | %-15s | %-15s\n",
    row$s,
    paste0(row$M_lambda_max, " (", row$RSD_lambda_max, ")"),
    paste0(row$M_MMLE_rank, " (", row$RSD_MMLE_rank, ")"),
    paste0(row$SIS_MLR, " (", row$RSD_MLR, ")"),
    paste0(row$SIS_MMLE, " (", row$RSD_MMLE, ")"), # Result from SIS package + MMS calc
    paste0(row$LASSO, " (", row$RSD_LASSO, ")"),
    paste0(row$SCAD, " (", row$RSD_SCAD, ")")
  ))
}
```

4 Evaluation

The simulation, designed to replicated Setting S3 from logistic regression model from Fan & Song(2010), finished executing 200 runs per sparsity level s . The final results are as below, which includes the $\mathcal{M} \cdot \lambda_{max}$ and the Minimum Model Size (MMS) statistics for the aforementioned methods, along with their respective Robust Standard Deviations (RSD).

TABLE 5 (Reproduced - Setting S3)
The Median MMS / M-lambda_max and the associated RSD (in parentheses)
Setting 3 (S3), $p = 2000$, $n = 600$, $\text{num_simulations} = 200$

s	M-lambda_max(RSD)	MMLE-Rank(RSD)	SIS-MLR(RSD)	SIS-MMLE(RSD)	LASSO(RSD)	SCAD(RSD)
3	11.63 (0.33)	3 (0)	3 (0)	51 (2.97)	16.5 (18.72)	5 (3.71)
6	17.21 (0.62)	6 (0)	6 (0)	91 (0.74)	61 (29.28)	6 (4.45)
12	28.8 (0.86)	23 (8.9)	24 (8.9)	55 (0)	59.5 (22.61)	39 (5.93)
24	52.33 (0.2)	1991 (8.15)	2000 (0)	67 (0)	2000 (0)	2000 (0)

When comparing these results to Table 5 in Fan & Song (2010) paper, several findings are worth pointing out:

- The majority of the recorded results are different from the numbers presented in the original study.
- The $\mathcal{M} \cdot \lambda_{max}$ values calculated in this simulation (11.63 to 52.33) is consistent higher than those reported in the paper (8.47 to 23.70). The trend of increasing $\mathcal{M} \cdot \lambda_{max}$ with s remains similar, yet higher magnitude suggest the generated data in this report might possess stronger collinearity or numeric properties than of the original study.
- The manual SIS implementation of MMLE-Rank, and SIS-MLR showed perfect performance in terms of low MMS metric and zero variance at sparsity level $s=3, 6$, and 12.
- At $s=12$, MMLE-Rank and SIS-MLR continued to outperform LASSO and SCAD, scoring at 23 and 24 model size while LASSO and SCAD needed 59.5 and 39 variables to recover the important information.
- The most noticeable deviation from the original results is that MMLE-Rank and SIS-MLR completely failed at high sparsity level $s=24$, with SIS-MLR unable to recover important variables and MMLE-Rank close to failure.
- A major discrepancy is while SCAD performs remarkably well for $s=3,6,12$ with recorded MMMS = 5,6,39 and LASSO performs adequately with recorded MMMS = 16.5, 61, and 59.5 in the original simulation, contrasting greatly the reported statistics in the original paper where they both failed under the same setting S3.
- All methods except SIS-MMLE failed to recover important variable in the simulation under $s=24$, with the failure patterns of LASSO and SCAD matches the original report.
- Another consistency is the effectiveness of SIS when using SIS function from R package under high sparsity setting $s=24$, where similarly to the original study, SIS-MMLE is able to retain relevant variables and LASSO, SCAD completely failed.

Overall, while total replication of the results were not achieved for Table 5, the simulation confirms the complexity of setting S3 and mostly show similar performance patterns than those reported by Fan & Song (2010). The failure of manual SIS at $s=24$ and the relative success of LASSO and SCAD at $s=6, 12$ are noticeable differences. Plus, it is also recorded that the SIS function from R package outperformed any other methods completely under high sparsity setting.

5 Conclusions

This report was an attempt to reproduce the simulation results for setting S3 from Table 5 of Fan & Song (2010) using R language. The report evaluated the performance of the extended Sure Independence Screening framework based on Marginal Likelihood Ratio (SIS-MLR) and Maximum Marginal Likelihood Estimates (SIS-MMLE). This report further explores the advantages of using SIS under certain conditions as it compared the performance metrics against proven methods such as LASSO and SCAD. The performance was primarily evaluated using the Minimum Model Size metric. To align more with the study by Fan & Song, the simulation strictly followed the data generation setup and the base model configurations presented in the paper. Due to the undisclosed coefficient magnitudes of β vector for Table 5, the study attempted to estimate this configuration using the setup in Table 4.

The resulting discrepancies between the two simulations are expected due to the unspecified parameters of true β^* coefficients which was used in the original study. The differences in environment and earlier version of R packages might also contributed to the differences.

Despite failing to reproduce the exact quantitative results of the original table, this study provides insights into the behavior of these variable selection methods in solving NP-dimensional problem. The results emphasize the promised potential in efficiency of SIS framework in effectively conducting variable selection in high sparsity and correlation level between the predictors. Through all levels of sparsity, the simulation shows that it completely outperformed LASSO and SCAD by being able to retain all the important information while recovering the least number of variables. Overall, the study reinforces the value of SIS as an efficient screening tool which is consistently more reliable in high-sparsity setting with complex correlation system between predictors. However, it is worth noticing that the report evaluated methods under certain conditions, specifically it is only predicated on logistic regression. Given the other simulation results in Fan & Song (2010), methods such as LASSO and SCAD remain more superior in linear regression setting with low sparsity level.

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